Q7

Answers written in the margins will not b

(c) In terms of contact area, explain why a 'full body harness' is less likely to cause injuries to or detac from the bungee jumper than a simple 'ankle attachment' during a fall. (2 marks

Answers written in the margins will not be marked.

7.

Figure 7.1

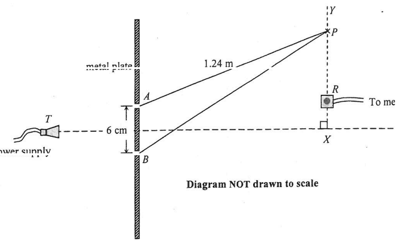

Figure 7.1 shows a set-up for the study of interference of microwaves. Microwaves of wavelength 2 emitted from a transmitter T pass through two slits A and B formed by metal plates. The slits are separa by 6 cm as shown. A probe R connecting to a meter is moved from X to Y to detect the intensity microwaves received. Transmitter T and point X are equidistant from A and B.

(a) Calculate the frequency of the microwaves. (2 marl

(b) (i) The meter shows alternate maxima and minima when R moves along XY. Explain. (2 marl

Answers written in the margins will not be marked.

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(b) (ii) The second minimum is found at position P where AP = 1.24 m. Find BP. (2 marl

(iii) When R is moved along XY from X towards Y and beyond, explain whether or not it is possit to detect more than three maxima. (2 mark

(c) Microwaves can be used in radar. Why are radio waves of lower frequencies not suitable for use radar ? (2 mark

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7. *(a) A laser light beam of wavelength 650 nm is incident normally on a pair of slits separated by $a = 0.325$ mm. Interference pattern is observed on a screen at a distance $D = 3.0$ m from the slits as shown in Figure 7.1. What is the separation between adjacent first- and second-order bright fringes ? (2 marks)

Figure 7.1

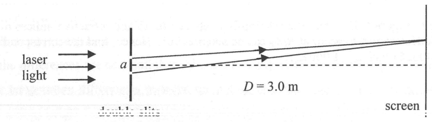

Diagram **NOT** drawn to scale

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(b) Figure 7.2 shows a set-up with two small LEDs separated by 6 mm and both LEDs emit light of wavelength 650 nm. State and explain what you would expect to see on the screen. (2 marks)

Figure 7.2

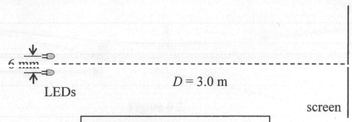

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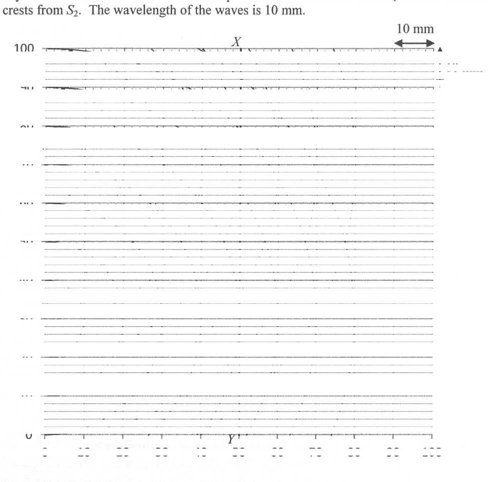
Figure 7.3 shows circular water waves in a ripple tank. The two point sources S_1 and S_2, separated by 40 mm, are driven by the same vibrator. The solid lines represent the wave crests from S_1 and the dotted lines represent the wave crests from S_2. The wavelength of the waves is 10 mm.



(c) Sketch on Figure 7.3 two lines to indicate all points P with path difference $PS_1 - PS_2$ equals to 10 mm (L_1) and 20 mm (L_2). State the kind of interference that occurs at these points P. (3 marks)

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(d) (i) If the interference pattern is observed along line XY at 50 mm from the sources as shown, **measure** the separation between adjacent first- and second-order maxima Δy. (1 mark)

separation Δy = ____________________

*(ii) However, using the calculation method in (a) would obtain 12.5 mm for this separation. Why does this calculated value differ with the measurement in (d)(i) ? (2 marks)

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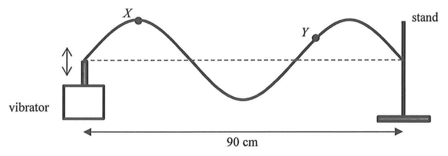
Q10

10. (a) A student uses the set-up in Figure 10.1 to investigate stationary waves on a stretched string. When the frequency of the vibrator is set at 100 Hz, a stationary wave is formed between the vibrator and the stand, which are 90 cm apart.

Figure 10.1



Particles X and Y on the string are at maximum displacement from their respective equilibrium positions at the moment shown in Figure 10.1.

(i) Label the positions of all antinodes (using 'A') and nodes (using 'N') on Figure 10.1. (2 marks)

(ii) Compare the oscillations of X and Y in terms of their amplitude and phase. (2 marks)

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(iii) Sketch on Figure 10.1 the shape of the string just after 0.0075 s. (1 mark)

(iv) If the frequency of the vibrator is increased gradually, at what frequency will the next stationary wave pattern be observed ? Show your working. (2 marks)

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Answers written in the margins will not be marked.

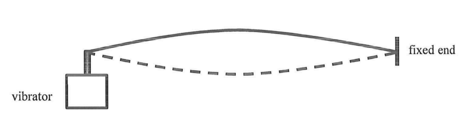
(b) Read the following passage about **stationary waves in musical instruments** and answer the questions that follow.

How do we make musical sounds ? In non-electronic instruments such as guitar and piano, stable and controlled vibration is usually produced by a stationary wave. If you pluck a guitar string fixed at both ends, various modes of stationary waves will be produced and they all travel with the same speed on the string. The mode with the lowest frequency f_1 is called the fundamental while the n^{th} mode, called the n^{th} harmonic, has a frequency n times that of the fundamental, i.e. nf_1.

The pitch of a note is determined by the frequency of the fundamental. The relative strengths and proportions of the harmonics determine the unique sound characteristics of different instruments. Therefore, a guitar and a violin playing a note of the same pitch will have different qualities because of different amplitudes and distributions of their harmonics.

(i) Figure 10.2 shows the stationary wave pattern of the fundamental formed on a string. Sketch the stationary wave pattern corresponding to the 2nd harmonic on Figure 10.2. (1 mark)

Figure 10.2



(ii) Explain why different musical instruments give different **sound qualities**. (2 marks)

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(iii) State **TWO** main differences between the **waves on a vibrating guitar string** and the **sound waves produced by it** besides the two waves being different in wavelength and speed. (2 marks)

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Answers written in the margins will not be marked.

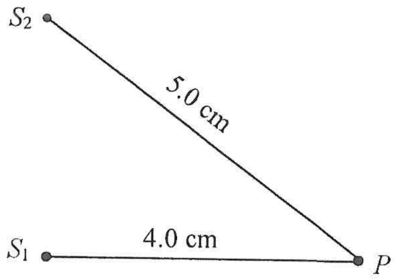
5. In a ripple tank, circular water waves are produced by two dippers S_1 and S_2 connected to a vibrator. The vibrator works in a way that S_1 and S_2 always vibrate in opposite directions as shown in Figure 5.1. The frequency of the vibration is 10 Hz and the wavelength λ of the waves is 2.0 cm.

Figure 5.1

(a) Find the wave speed of the waves. (1 mark)

(b) (i)

Figure 5.2



P is a point at the water surface as shown in Figure 5.2. Express the path difference at P from S_1 and S_2 in terms of λ. (2 marks)

(ii) Explain briefly what type of interference occurs at P. (2 marks)

(c) If the depth of the water in the ripple tank is increased while other conditions remain unchanged, deduce how the wavelength of the waves will be affected. (2 marks)